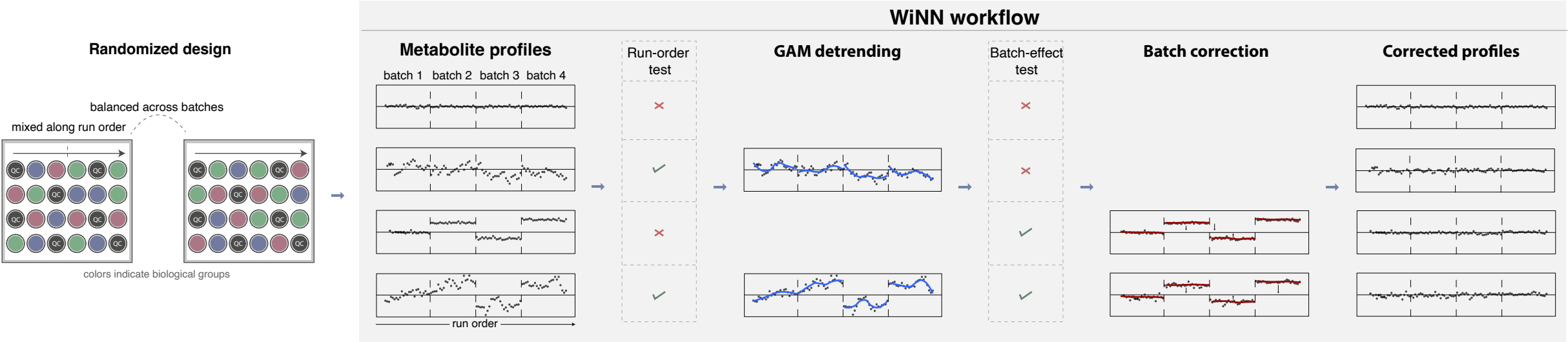
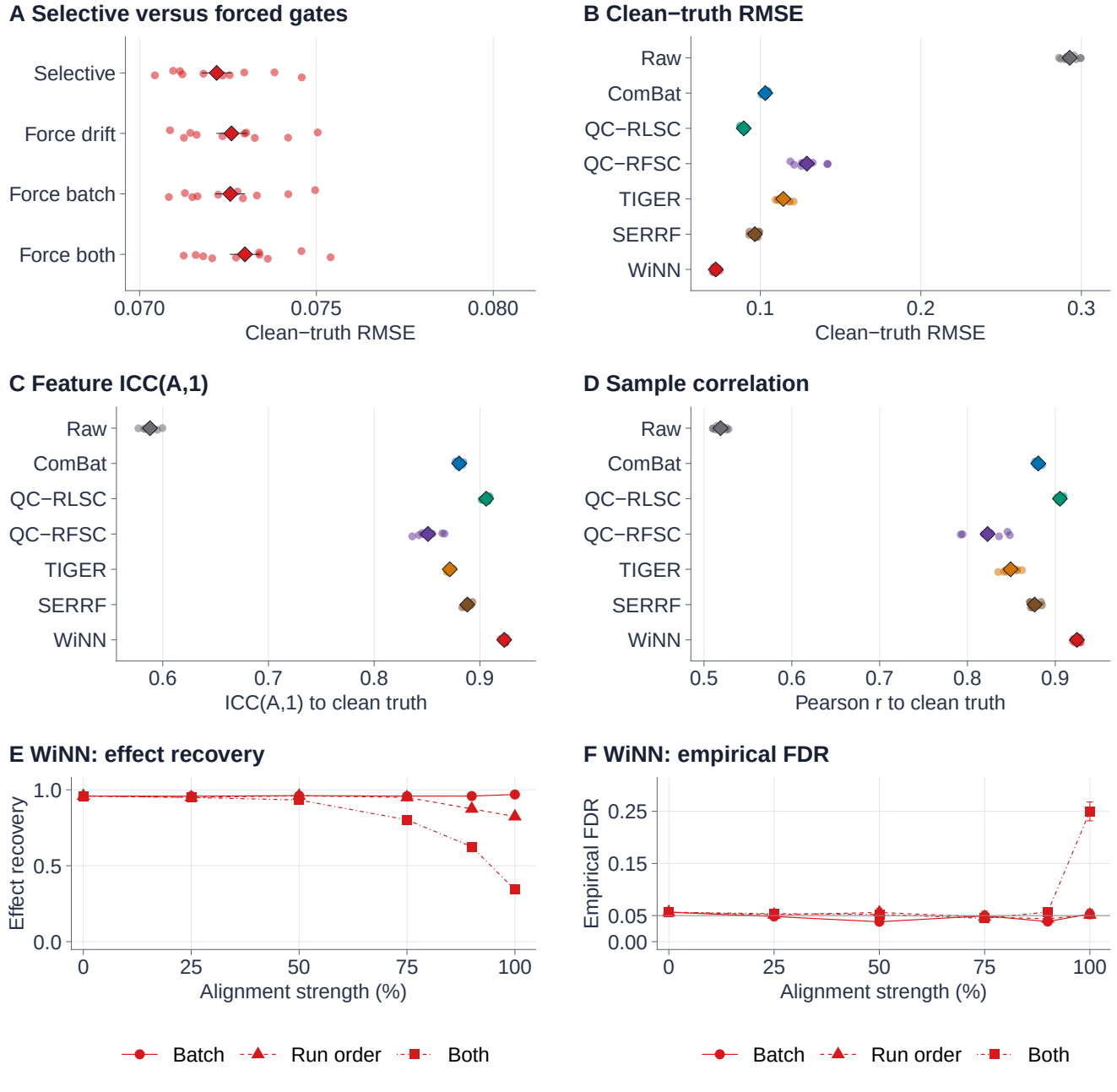
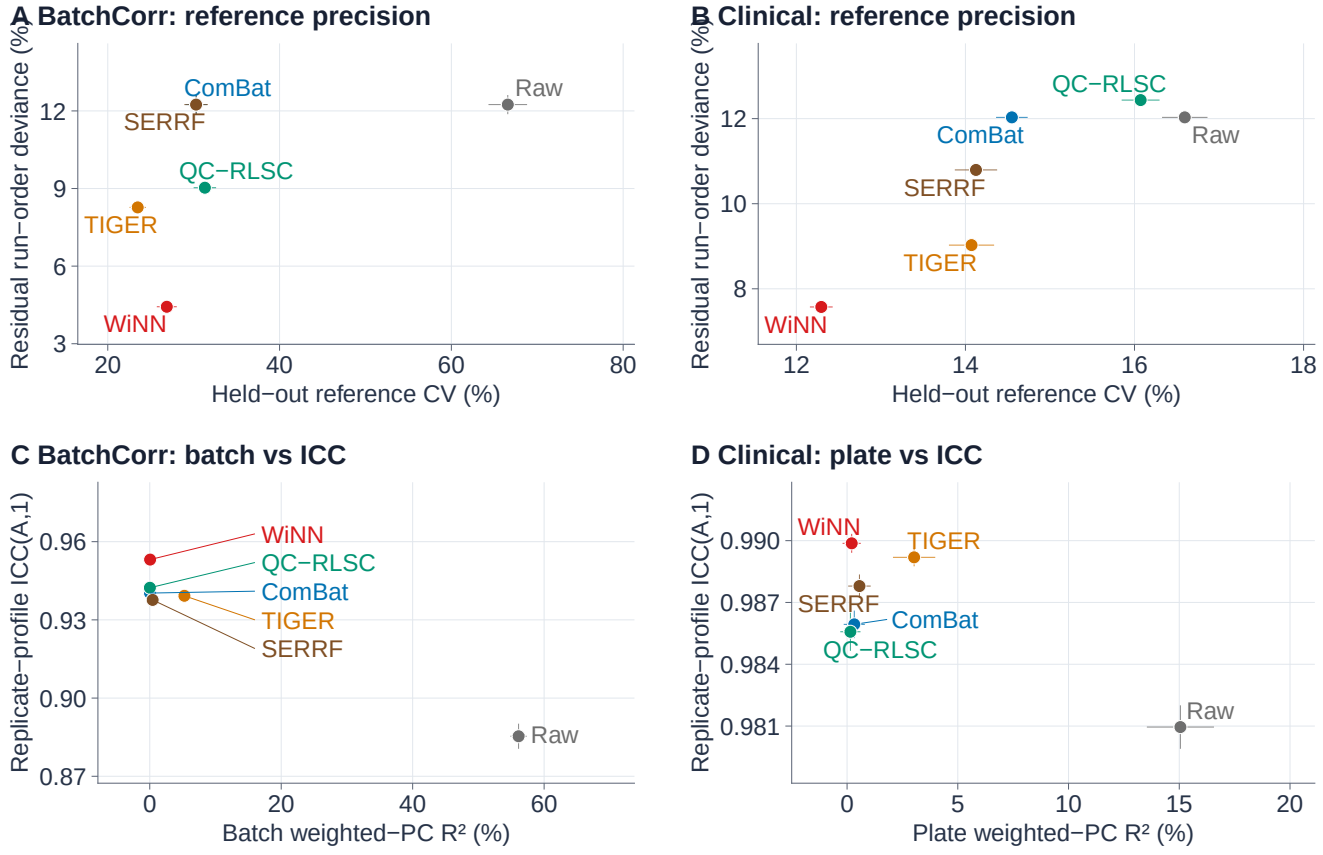
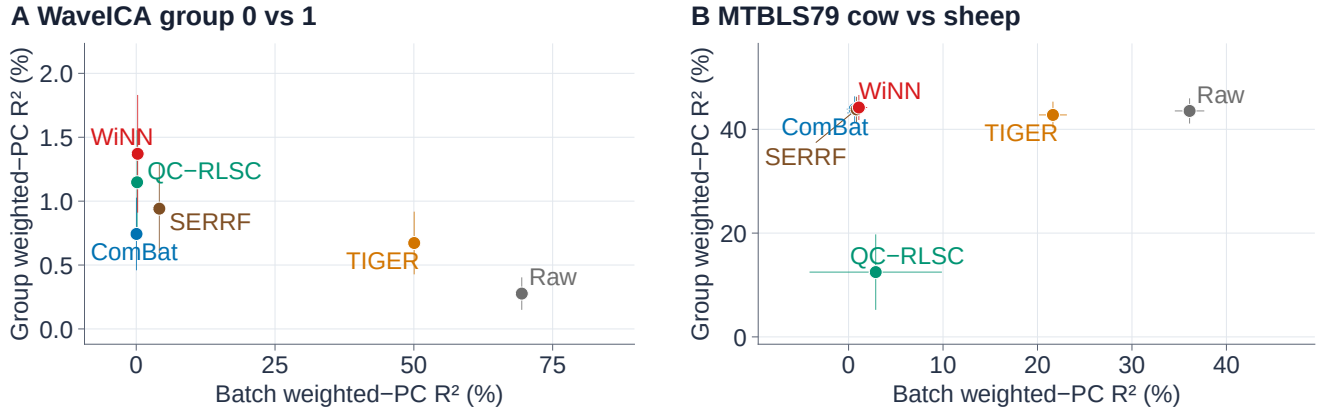

Randomized design
balanced across batches
mixed along run order
QC
colors indicate biological groups
WiNN workflow
Metabolite profiles
batch 1
batch 2
batch 3
batch 4
run order
Run-order test
GAM detrending
Batch-effect test
Batch correction
Corrected profiles


A Selective versus forced gates
Selective
Force drift
Force batch
Force both
0.070
0.075
0.080
Clean−truth RMSE
B Clean−truth RMSE
Raw
ComBat
QC−RLSC
QC−RFSC
TIGER
SERRF
WiNN
0.1
0.2
0.3
Clean−truth RMSE
C Feature ICC(A,1)
0.6
0.7
0.8
0.9
ICC(A,1) to clean truth
D Sample correlation
0.5
0.6
0.7
0.8
0.9
Pearson r to clean truth
E WiNN: effect recovery
Effect recovery
1.0
0.5
0.0
0
25
50
75
100
Alignment strength (%)
F WiNN: empirical FDR
Empirical FDR
0.25
0.15
0.05
0.00
0
25
50
75
100
Alignment strength (%)
Batch
Run order
Both
Batch
Run order
Both


A BatchCorr: reference precision
Residual run−order deviance (%)
Held−out reference CV (%)
ComBat
Raw
SERRF
QC−RLSC
TIGER
WiNN
B Clinical: reference precision
Residual run−order deviance (%)
Held−out reference CV (%)
QC−RLSC
ComBat
Raw
SERRF
TIGER
WiNN
C BatchCorr: batch vs ICC
Replicate−profile ICC(A,1)
Batch weighted−PC R² (%)
WiNN
QC−RLSC
ComBat
TIGER
SERRF
Raw
D Clinical: plate vs ICC
Replicate−profile ICC(A,1)
Plate weighted−PC R² (%)
WiNN
TIGER
SERRF
ComBat
QC−RLSC
Raw


A WaveICA group 0 vs 1
Group weighted−PC R² (%)
Batch weighted−PC R² (%)
2.0
1.5
1.0
0.5
0.0
0
25
50
75
WiNN
QC−RLSC
SERRF
ComBat
TIGER
Raw
B MTBLS79 cow vs sheep
Group weighted−PC R² (%)
Batch weighted−PC R² (%)
40
20
0
0
10
20
30
40
WiNN
ComBat
SERRF
TIGER
Raw
QC−RLSC